

Discovery of a magnetic field in the O-type bright giant 63 Oph

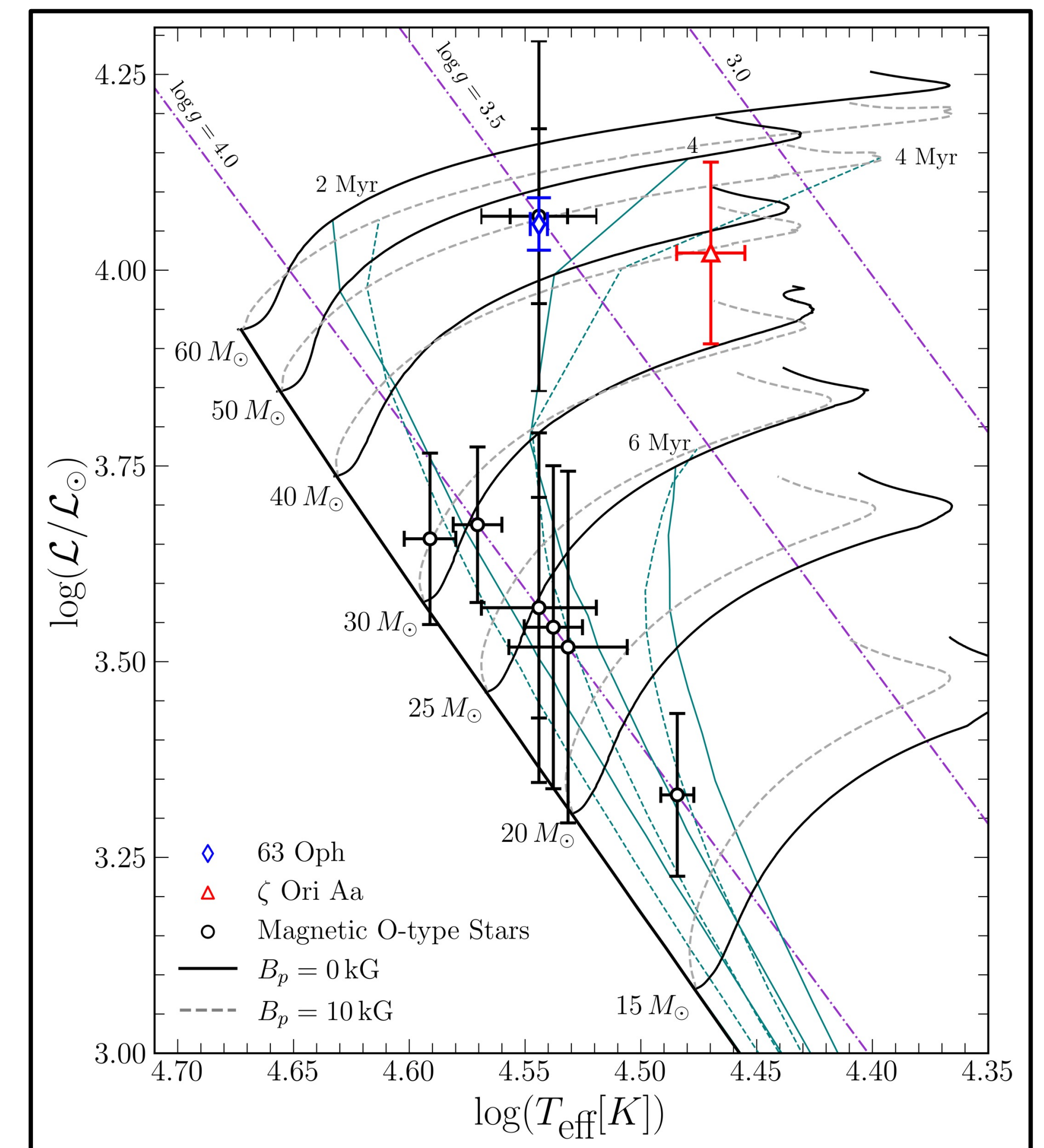
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Magnetic massive stars

- Surface magnetic fields are detected in fewer than 10% of O-type stars and are even rarer in those near the terminal age main sequence (TAMS) [1, 2].
- It remains unclear if the rarity of magnetic massive stars near the TAMS is due to **detection biases** or **magnetic field decay** [2].
- Magnetic fields impact O star evolutionary paths by slowing rotation through magnetic braking and by reducing their mass loss [e.g., 3, 4, 5].

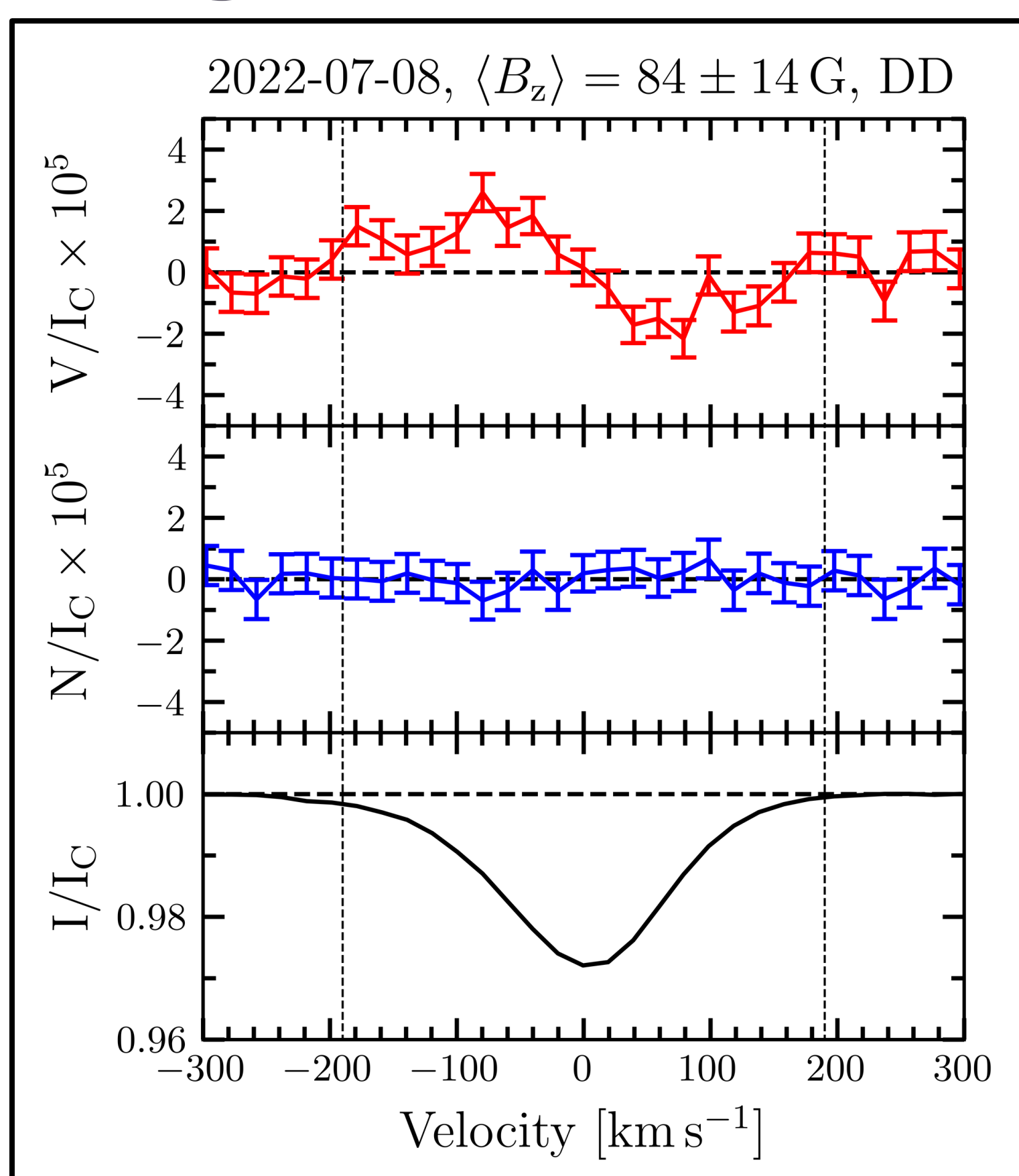
63 Oph: O8 II((f)), $T_{\text{eff}} \approx 35$ kK, $\log g \approx 3.5$, $V_{\text{ini}} \approx 50$ km/s, $V_{\text{mac}} \approx 100$ km/s

- Originally identified as a magnetic candidate in the MiMeS survey [1, 2].
- Three spectropolarimetric observations were obtained using ESPaDOnS at the Canada-France-Hawaii Telescope to search for a magnetic signature.
- Least-squares deconvolution [6] was applied to produce mean Stokes I (unpolarized), Stokes V (circularly polarized), and null (N) profiles.
- Variability analysis was conducted on 30 additional FEROS, FIES and HERMES spectra and archival Hipparcos, ASAS-SN and K2 photometry.

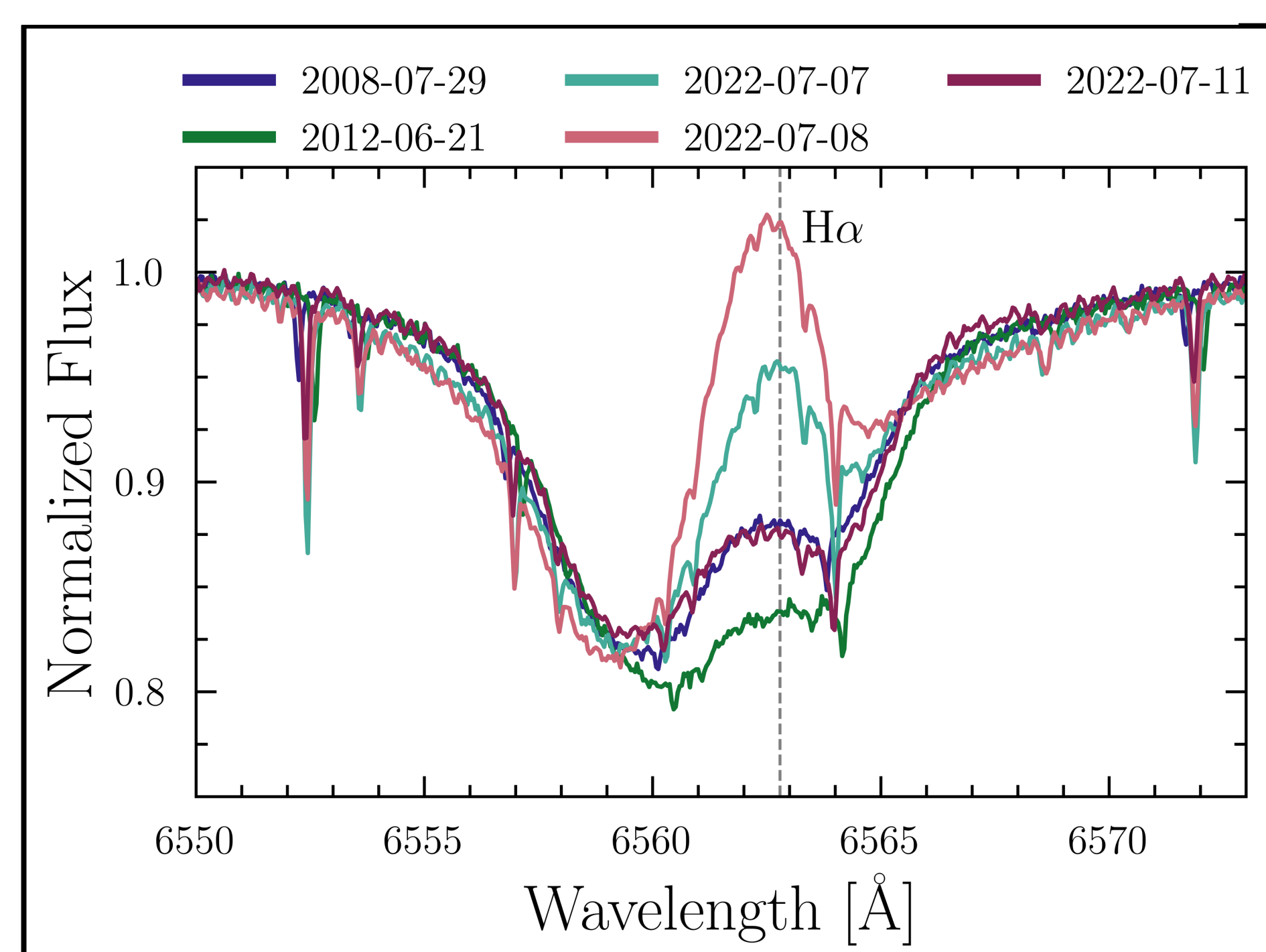


MESA magnetic tracks from [5].

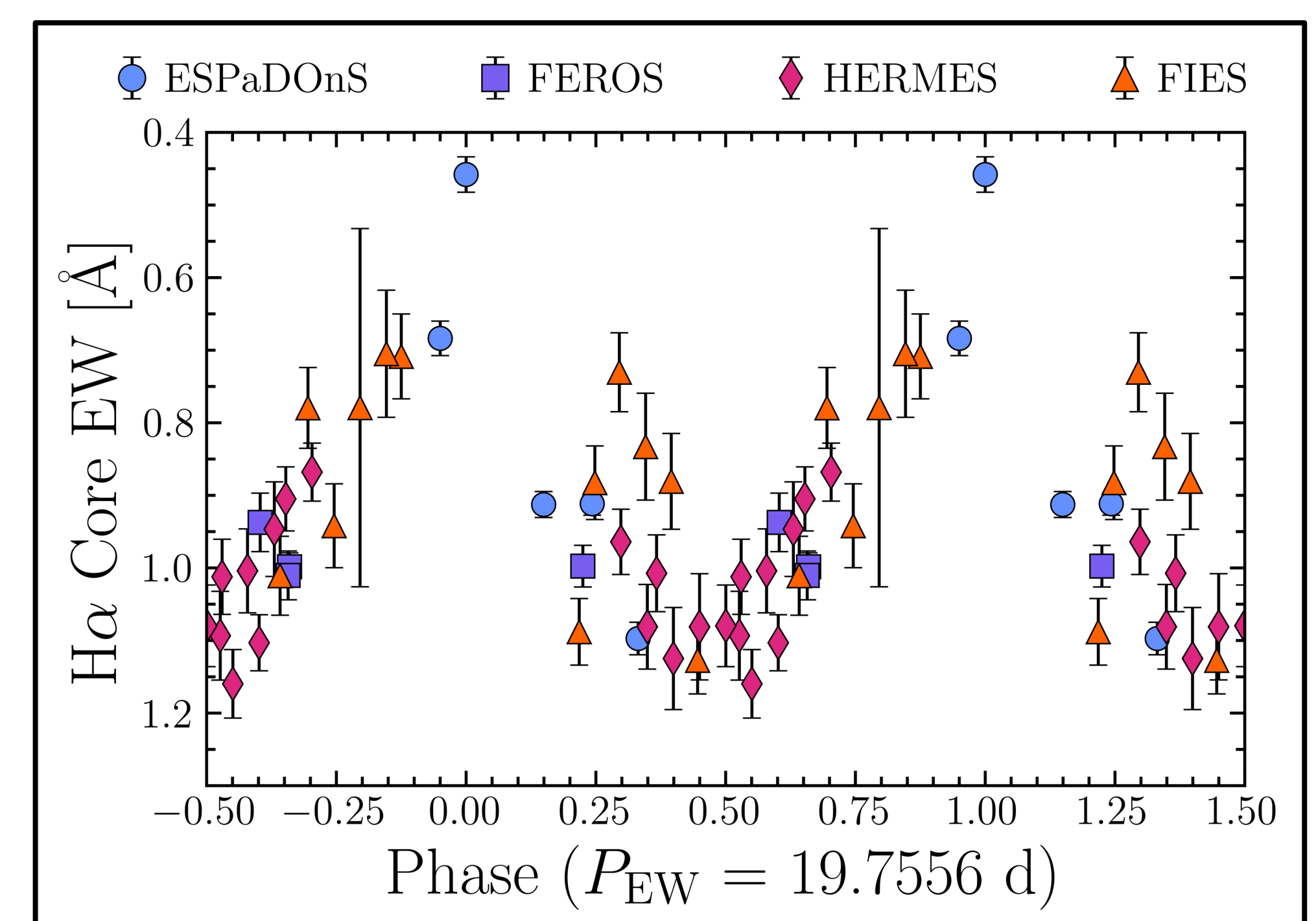
Magnetic detection and spectral variability



A magnetic signature is **detected** in one LSD Stokes V profile (red line). The longitudinal magnetic field is **84 ± 14 G**.



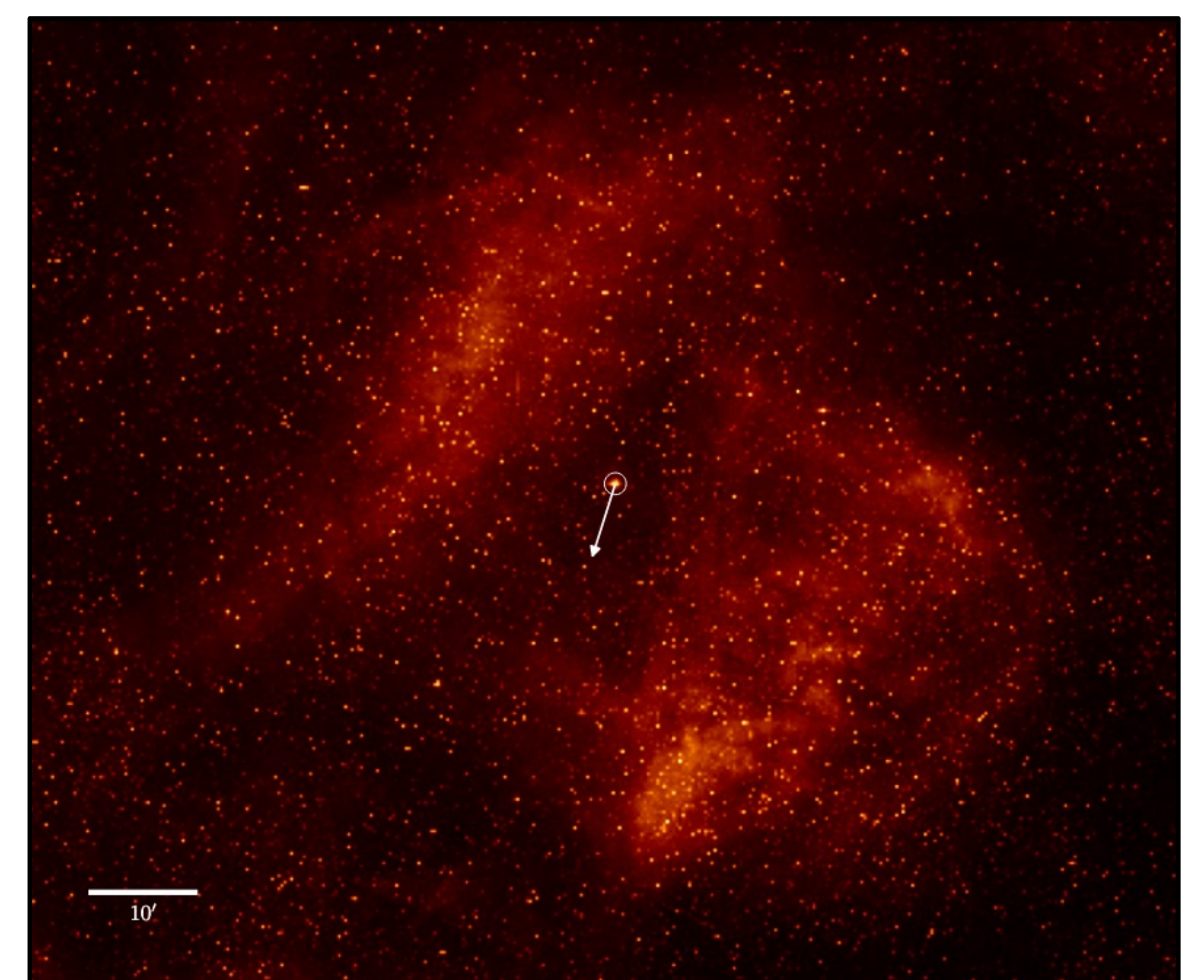
Variability of the **H α** line in ESPaDOnS spectra. The peak emission corresponds to the magnetic detection.



H α line-core equivalent widths phased to the best fitting period of **≈ 19.8 d**. We interpret the variability as modulation due to a **rotating magnetosphere**.

Conclusions and future work

- 63 Oph may be a **rare link** between the **strongly magnetic** O-type stars detected near the zero-age main-sequence and **weakly magnetic** O-type supergiants such as ζ Ori Aa.
- Assuming a dipolar magnetic field geometry, we place a lower limit on the dipole field strength of **$B_p \geq 300 \pm 50$ G**.
- No evidence for a binary companion in RVs or time series photometry.
- Additional spectropolarimetric observations are required to constrain the magnetic field strength and geometry and enable detailed evolutionary modeling.
- The sample of IACOB O-type stars with similar spectral characteristics (quality flag Q3) should be further investigated for hidden, evolved magnetic stars like 63 Oph [7].
- **Results are presented in Barron et al. 2025 (ApJ, in press; DOI: 10.3847/1538-4357/adf6d9; arXiv: 2505.11374).**



Study of the surrounding **nebula** may provide insight into 63 Oph's history and help test the **merger hypothesis** for the origin of magnetic massive stars.

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